beijing

Full twelve-station hourly panel. Supplied NA measurements remain missing cells, distinct from absent records.

Completed evidence · 1 sources · 39 retained records

15 evidence-qualified finding(s) across 1 supplied source(s). Required stages completed over eligible records; behavioral search scope is disclosed separately.

Actions to investigate

Window effects meet configured investigation settings

Review reference support, dispersion, seasonality, and entity composition before interpreting the change. Review the declarations and bounded witnesses.

investigation_candidate · t7 · PM2.5

Window effects meet configured investigation settings

Review reference support, dispersion, seasonality, and entity composition before interpreting the change. Review the declarations and bounded witnesses.

investigation_candidate · t23 · DEWP

Contiguous equal-value runs merit investigation

Configured candidate threshold met. Legitimate plateaus, zeros, and operating regimes remain plausible. Review the declarations and bounded witnesses.

investigation_candidate · t29 · RAIN

Window effects meet configured investigation settings

Review reference support, dispersion, seasonality, and entity composition before interpreting the change. Review the declarations and bounded witnesses.

investigation_candidate · t31 · WSPM

Contiguous equal-value runs merit investigation

Configured candidate threshold met. Legitimate plateaus, zeros, and operating regimes remain plausible. Review the declarations and bounded witnesses.

investigation_candidate · t33 · WSPM

Interpretation boundary

- No composite score or universal readiness verdict.
- Investigation defaults are not validated defect criteria.
- Supplied evidence does not prove what a production pipeline consumed.

Generated 2026-10-10T21:16:46.186667+00:00 · Schema 1.4 · temporal_reliability_v1

Declarations

Evidence / source	Measurement or rule	Retained result
beijing	Source and base task	{"format":"csv","role":"dataset","task":"dataset_profile","targets":null}
beijing	observation	measurement
beijing	layout	wide
beijing	entity columns	["station"]
beijing	observation keys	[]
beijing	clock	{"column":"time","kind":"instant","encoding":"iso","format":null,"epoch_unit":null,"timezone":"Asia/Shanghai","timezone_column":null,"ambiguity":"unresolved","fold_column":null,"date_convention":null}
beijing	schedule	{"kind":"fixed","cadence_ns":"3600000000000","anchor":"2013-03-01T00:00:00+08:00","civil_unit":null,"timezone":null,"month_end":"skip","source_id":null,"time_column":null,"entity_columns":[],"measurement_column":null,"expectation":"required_record","collection_status_column":null,"collection_active_values":["active"]}
beijing	active intervals	[{"start":"2013-03-01T00:00:00+08:00","end":"2017-03-01T00:00:00+08:00","entity":[],"measurement":null,"closed":"left"}]
beijing	PM2.5	{"name":"PM2.5","column":null,"representation":"quantitative","unit":"ug/m3","meaning":null,"past_observation":false}
beijing	PM10	{"name":"PM10","column":null,"representation":"quantitative","unit":"ug/m3","meaning":null,"past_observation":false}
beijing	TEMP	{"name":"TEMP","column":null,"representation":"quantitative","unit":null,"meaning":null,"past_observation":false}
beijing	PRES	{"name":"PRES","column":null,"representation":"quantitative","unit":null,"meaning":null,"past_observation":false}
beijing	DEWP	{"name":"DEWP","column":null,"representation":"quantitative","unit":null,"meaning":null,"past_observation":false}
beijing	RAIN	{"name":"RAIN","column":null,"representation":"quantitative","unit":null,"meaning":null,"past_observation":false}
beijing	WSPM	{"name":"WSPM","column":null,"representation":"quantitative","unit":null,"meaning":null,"past_observation":false}
beijing	wd	{"name":"wd","column":null,"representation":"categorical","unit":null,"meaning":null,"past_observation":false}

16 of 16 metric rows shown. Presentation limits do not change analytical coverage.

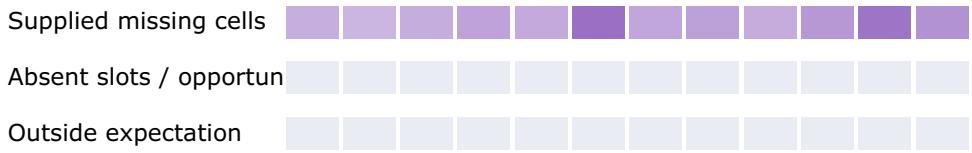
Chronology

Evidence / source	Measurement or rule	Retained result
t1 beijing	rows	420768
t1 beijing	cells	3366144
t1 beijing	declared dense missing cells	0
t1 beijing	parse states / valid	3366144
t1 beijing	precision counts / second	3366144
t1 beijing	value states [DEWP] / measurement	DEWP
t1 beijing	value states [DEWP] / state	missing
t1 beijing	value states [DEWP] / cells	403
t1 beijing	value states [DEWP] / measurement	DEWP
t1 beijing	value states [DEWP] / state	valid
t1 beijing	value states [DEWP] / cells	420365
t1 beijing	value states [PM10] / measurement	PM10
t1 beijing	value states [PM10] / state	missing
t1 beijing	value states [PM10] / cells	6449
t1 beijing	value states [PM10] / measurement	PM10
t1 beijing	value states [PM10] / state	valid
t1 beijing	value states [PM10] / cells	414319
t1 beijing	value states [PM2.5] / measurement	PM2.5
t1 beijing	value states [PM2.5] / state	missing
t1 beijing	value states [PM2.5] / cells	8739

20 of 69 metric rows shown. 49 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

Continuity and history

Absent observations, supplied missing values and outside-expectation periods have different meanings. Event absence alone does not prove an outage.



beijing · missing values and absent slots remain separate



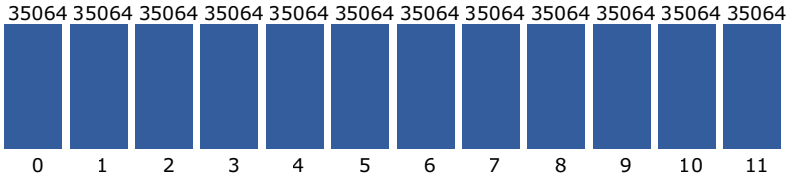
beijing · expected population

Evidence / source	Measurement or rule	Retained result
t4 beijing	distinct entities	12
t4 beijing	distinct series	96
t4 beijing	history counts / min	35064
t4 beijing	history counts / max	35064
t4 beijing	history counts / entity weighted mean	35064
t4 beijing	history length bins [0] / observations	4
t4 beijing	history length bins [0] / series	96
t4 beijing	observed first tick	1362067200000000000
t4 beijing	observed last tick	1488294000000000000
t4 beijing	observed entry exit is lifecycle	No
t4 beijing	retained histories [DEWP] / entity	["Aotizhongxin"]
t4 beijing	retained histories [DEWP] / measurement	DEWP
t4 beijing	retained histories [DEWP] / observations	35064
t4 beijing	retained histories [DEWP] / cells	35064
t4 beijing	retained histories [DEWP] / first tick	1362067200000000000
t4 beijing	retained histories [DEWP] / last tick	1488294000000000000
t4 beijing	retained histories [DEWP] / span ticks	126226800000000000
t4 beijing	retained histories [PM10] / entity	["Aotizhongxin"]
t4 beijing	retained histories [PM10] / measurement	PM10
t4 beijing	retained histories [PM10] / observations	35064

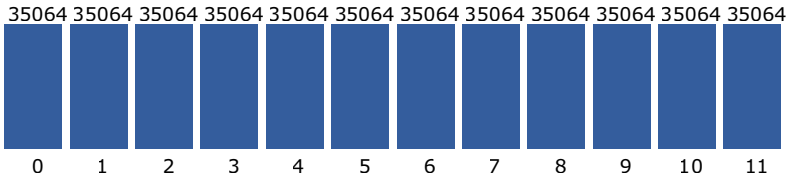
20 of 818 metric rows shown. 798 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

Behavioral investigation

Thresholds and support settings are configurable investigation defaults. Reference support, sparse windows, zero dispersion, phase exclusions and population mix qualify interpretation.



beijing / PM2.5 · supplied cells per window



beijing / PM10 · supplied cells per window

Evidence / source	Measurement or rule	Retained result
t7 beijing / PM2.5	Reference and interpretation	{"reference_window":0,"unit":"ug/m3","meaning":"Observed effects are descriptive investigation evidence, not validated universal defects, changepoints, concept drift, or model performance."}
t7 beijing / PM2.5	Window 0	{"window_id":0,"start_tick":"1362067200000000000","end_tick":"1372586100000000000","cells":35064,"missing":645,"invalid":0,"nonfinite":0,"finite":34419,"entities":12,"missing_rate":0.018394934976043806,"quantiles":[3.0,7.0,32.0,70.0,124.0,230.0,326.0],"iqr":92.0}
t7 beijing / PM2.5	Window 1	{"window_id":1,"start_tick":"1372586100000000000","end_tick":"1383105000000000000","cells":35064,"missing":525,"invalid":0,"nonfinite":0,"finite":34539,"entities":12,"missing_rate":0.014972621492128678,"quantiles":[3.0,7.0,25.0,54.0,102.0,214.0,291.6200000000026],"iqr":77.0}
t7 beijing / PM2.5	Window 2	{"window_id":2,"start_tick":"1383105000000000000","end_tick":"1393623900000000000","cells":35064,"missing":583,"invalid":0,"nonfinite":0,"finite":34481,"entities":12,"missing_rate":0.01662673967602099,"quantiles":[3.0,6.0,19.0,69.0,144.0,300.0,445.0],"iqr":125.0}
t7 beijing / PM2.5	Window 3	{"window_id":3,"start_tick":"1393623900000000000","end_tick":"1404142800000000000","cells":35064,"missing":766,"invalid":0,"nonfinite":0,"finite":34298,"entities":12,"missing_rate":0.021845767738991557,"quantiles":[3.0,9.0,27.0,58.0,106.0,201.0,286.0],"iqr":79.0}
t7 beijing / PM2.5	Window 4	{"window_id":4,"start_tick":"1404142800000000000","end_tick":"1414661700000000000","cells":35064,"missing":765,"invalid":0,"nonfinite":0,"finite":34299,"entities":12,"missing_rate":0.021817248459958933,"quantiles":[3.0,7.0,28.0,65.0,117.0,235.0,340.0],"iqr":89.0}
t7 beijing / PM2.5	Window 5	{"window_id":5,"start_tick":"1414661700000000000","end_tick":"1425180600000000000","cells":35064,"missing":968,"invalid":0,"nonfinite":0,"finite":34096,"entities":12,"missing_rate":0.027606662103582022,"quantiles":[3.0,5.0,13.0,51.0,130.0,265.0,360.0],"iqr":117.0}
t7 beijing / PM2.5	Window 6	{"window_id":6,"start_tick":"1425180600000000000","end_tick":"1435699500000000000","cells":35064,"missing":764,"invalid":0,"nonfinite":0,"finite":34300,"entities":12,"missing_rate":0.021788729180926305,"quantiles":[3.0,7.0,23.0,52.0,98.0,183.0,272.0],"iqr":75.0}
t7 beijing / PM2.5	Window 7	{"window_id":7,"start_tick":"1435699500000000000","end_tick":"1446218400000000000","cells":35064,"missing":823,"invalid":0,"nonfinite":0,"finite":34241,"entities":12,"missing_rate":0.023471366643851244,"quantiles":[3.0,6.0,17.0,38.0,76.0,166.0,311.0],"iqr":59.0}
t7 beijing / PM2.5	Window 8	{"window_id":8,"start_tick":"1446218400000000000","end_tick":"1456737300000000000","cells":35064,"missing":531,"invalid":0,"nonfinite":0,"finite":34533,"entities":12,"missing_rate":0.015143737166324436,"quantiles":[3.0,6.0,15.0,51.0,135.0,309.0,531.0],"iqr":120.0}
t7 beijing / PM2.5	Window 9	{"window_id":9,"start_tick":"1456737300000000000","end_tick":"1467256200000000000","cells":35064,"missing":921,"invalid":0,"nonfinite":0,"finite":34143,"entities":12,"missing_rate":0.026266255989048595,"quantiles":[3.0,7.0,22.0,49.0,88.0,208.0,335.0],"iqr":66.0}

Evidence / source	Measurement or rule	Retained result
t7 beijing / PM2.5	Window 10	{"window_id":10,"start_tick":"1467256200000000000","end_tick":"1477775100000000000","cells":35064,"missing":1072,"invalid":0,"nonfinite":0,"finite":33992,"entities":12,"missing_rate":0.03057266712297513,"quantiles":[3.0,7.0,22.0,48.0,92.0,171.0,244.0899999999965],"iqr":70.0}
t7 beijing / PM2.5	Window 11	{"window_id":11,"start_tick":"1477775100000000000","end_tick":"1488294000000000001","cells":35064,"missing":376,"invalid":0,"nonfinite":0,"finite":34688,"entities":12,"missing_rate":0.010723248916267396,"quantiles":[3.0,5.0,15.0,66.0,154.0,326.0,483.0],"iqr":139.0}
t7 beijing / PM2.5	Comparison 0 → 1	{"reference_window":0,"comparison_window":1,"reference_support":35064,"comparison_support":35064,"reference_iqr":92.0,"reference_span":["1362067200000000000","1372586100000000000"],"reference_entities":12,"comparison_entities":12,"unit":"ug/m3","thresholds_are_investigation_defaults":true,"missingness_change":-0.003422313483915128,"missingness_r… [retained JSON contains full detail]
t7 beijing / PM2.5	Comparison 1 → 2	{"reference_window":1,"comparison_window":2,"reference_support":35064,"comparison_support":35064,"reference_iqr":77.0,"reference_span":["1372586100000000000","1383105000000000000"],"reference_entities":12,"comparison_entities":12,"unit":"ug/m3","thresholds_are_investigation_defaults":true,"missingness_change":0.0016541181838923124,"missingness_r… [retained JSON contains full detail]
t7 beijing / PM2.5	Comparison 2 → 3	{"reference_window":2,"comparison_window":3,"reference_support":35064,"comparison_support":35064,"reference_iqr":125.0,"reference_span":["1383105000000000000","1393623900000000000"],"reference_entities":12,"comparison_entities":12,"unit":"ug/m3","thresholds_are_investigation_defaults":true,"missingness_change":0.005219028062970567,"missingness_r… [retained JSON contains full detail]
t7 beijing / PM2.5	Comparison 3 → 4	{"reference_window":3,"comparison_window":4,"reference_support":35064,"comparison_support":35064,"reference_iqr":79.0,"reference_span":["1393623900000000000","1404142800000000000"],"reference_entities":12,"comparison_entities":12,"unit":"ug/m3","thresholds_are_investigation_defaults":true,"missingness_change":-2.851927903262455e-05,"missingness_… [retained JSON contains full detail]
t7 beijing / PM2.5	Comparison 4 → 5	{"reference_window":4,"comparison_window":5,"reference_support":35064,"comparison_support":35064,"reference_iqr":89.0,"reference_span":["1404142800000000000","1414661700000000000"],"reference_entities":12,"comparison_entities":12,"unit":"ug/m3","thresholds_are_investigation_defaults":true,"missingness_change":0.005789413643623089,"missingness_re… [retained JSON contains full detail]
t7 beijing / PM2.5	Comparison 5 → 6	{"reference_window":5,"comparison_window":6,"reference_support":35064,"comparison_support":35064,"reference_iqr":117.0,"reference_span":["1414661700000000000","1425180600000000000"],"reference_entities":12,"comparison_entities":12,"unit":"ug/m3","thresholds_are_investigation_defaults":true,"missingness_change":-0.005817932922655717,"missingness_… [retained JSON contains full detail]
t7 beijing / PM2.5	Comparison 6 → 7	{"reference_window":6,"comparison_window":7,"reference_support":35064,"comparison_support":35064,"reference_iqr":75.0,"reference_span":["1425180600000000000","1435699500000000000"],"reference_entities":12,"comparison_entities":12,"unit":"ug/m3","thresholds_are_investigation_defaults":true,"missingness_change":0.0016826374629249387,"missingness_r… [retained JSON contains full detail]

20 of 5039 metric rows shown. 5019 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

RETAINED EVIDENCE · 6 / 8

Availability and outcome maturity

Undeclared availability remains unknown. Historical agreement concerns supplied coverage; it does not establish what a pipeline consumed.

Evidence / source	Measurement or rule	Retained result
t6 beijing	features	["PM2.5","PM10","TEMP","PRES","DEWP","RAIN","WSPM","wd"]
t6 beijing	meaning	Undeclared availability remains unknown; no timestamp inheritance.

2 of 2 metric rows shown. Presentation limits do not change analytical coverage.

RETAINED EVIDENCE · 7 / 8

Chronological evaluation

No retained evidence applies to this topic in the declared scope.

0 of 0 metric rows shown. Presentation limits do not change analytical coverage.

Methodology and coverage

Evidence / source	Measurement or rule	Retained result
method	schema	1.4
method	methodology	temporal_reliability_v1
method	execution status	completed
method	aggregate scope	Every eligible declared record in required stages; no silent sampling.
method	search scope [0] / selected measurements	["PM2.5","PM10","TEMP","PRES","DEWP","RAIN","WSPM","wd"]
method	search scope [0] / unsearched scope	[]
method	search scope [0] / work observation visits	65639808
method	search scope [0] / comparisons	840
method	search scope [0] / selection	Requested measurements first, then declaration order. Detail retention can miss behavior-only exceptions.
method	search scope [0] / defaults validated universal defect criteria	No
method	internal detail omissions / beijing:entity histories	0
method	internal detail omissions / beijing:series effects	236
method	timezone / package	tzdata
method	timezone / version	2026.5
method	timezone / zones / Asia/Shanghai	bf8b7ed82fe6e63e6d98f8cea934eeac901cd16aba85eb5755ce3f8b4289ea8a
method	phase seconds / beijing:ingestion	3.373883
method	phase seconds / beijing:normalization	5.526881
method	phase seconds / required diagnostics	3.698931
method	phase seconds / availability history evaluation	1.295801e-05
method	phase seconds / behavior	11.45997

20 of 34 metric rows shown. 14 presentation rows omitted; retained JSON preserves their evidence. Presentation limits do not change analytical coverage.

Decoded-batch controls bound accepted batches and application retention. Native pages, dictionaries and conversion buffers can allocate before checks. Estimates and workload limits are provisional engineering controls. RSS monitoring can overshoot between samples.

Analytical scope left unsearched, retained-detail omissions, and presentation omissions are separate. No independent-row significance testing, model fitting, split generation, automatic repair or universal temporal score is performed.

Retained witnesses and selected detail

100 of 236 retained detail rows available here. Print shows up to 20 rows; JSON preserves every retained record. These presentation omissions do not change searched scope.

Evidence	Detail	Witness / selected population
t4	retained histories	{"entity":["Aotizhongxin"],"measurement":"DEWP","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity":["Aotizhongxin"],"measurement":"PM10","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity":["Aotizhongxin"],"measurement":"PM2.5","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity":["Aotizhongxin"],"measurement":"PRES","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity":["Aotizhongxin"],"measurement":"RAIN","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity":["Aotizhongxin"],"measurement":"TEMP","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity":["Aotizhongxin"],"measurement":"WSPM","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity":["Aotizhongxin"],"measurement":"wd","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity":["Changping"],"measurement":"DEWP","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity":["Changping"],"measurement":"PM10","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity":["Changping"],"measurement":"PM2.5","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity":["Changping"],"measurement":"PRES","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity":["Changping"],"measurement":"RAIN","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity":["Changping"],"measurement":"TEMP","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity":["Changping"],"measurement":"WSPM","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}

Evidence	Detail	Witness / selected population
t4	retained histories	{"entity": ["Changping"],"measurement":"wd","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity": ["Dingling"],"measurement":"DEWP","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity": ["Dingling"],"measurement":"PM10","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity": ["Dingling"],"measurement":"PM2.5","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}
t4	retained histories	{"entity": ["Dingling"],"measurement":"PRES","observations":35064,"cells":35064,"first_tick":"1362067200000000000","last_tick":"1488294000000000000","span_ticks":"126226800000000000"}